

CMMS Creation, Configuration, and Transaction Page Documentation

Application: Solar Power Plant Computerized Maintenance Management System (CMMS)

Document version: 1.0

Verified against source: 21 July 2026

Audience: Business users, plant operators, maintenance and store managers, administrators, developers, testers, and client/demo users

Document Control and Scope

This document describes every currently implemented route or embedded UI workflow that creates, edits, configures, approves, assigns, uploads, issues, consumes, returns, transfers, or submits application data. Field names, APIs, permissions, and database mappings are based on the current frontend, backend, Liquibase, and `api-permission-mapping.csv`.

The application uses a standard JSON response envelope (`ApiResponse<T>`). API paths below include the configured `/api` servlet context. Site-filtered operations additionally apply record-level site access in backend services.

How to read field tables

- **UI/Required** identifies the visible control and whether the current UI requires it.
- **Technical mapping** lists frontend property → DTO property → database column.
- **Source/Edit** explains dropdown origin, defaults, dependencies, and edit restrictions.
- “Backend generated” means users do not enter the value.

Application Creation Order

Company profile

- Sites
- Roles and permissions
- Employees/login accounts
- Vendors and AMC contracts
- Equipment
- PM schedules and maintenance requests
- Assignments
- Checklists, work logs, downtime, and spare requests
- Stock fulfilment, transfers, reorders, consumption, and reports

Actual Role and Permission Model

Routes and buttons use permission codes rather than hard-coded business-role names. Roles are configurable; therefore “who uses this page” means users whose assigned role contains the stated permission. Common seeded/business-facing role codes referenced by current configuration include `SUPER_ADMIN`, `ADMIN`, `MAINTENANCE_MANAGER`, and `EMPLOYEE`, but clients can create additional roles.

1. Company Profile Create/Edit

Purpose, users, and timing

Navigation: Administration → Company

Route: /admin/company

Users: Company administrators with `COMPANY_VIEW`; saving requires `COMPANY_CREATE` for the first record or `COMPANY_UPDATE` afterward.

Use when: initially configuring the tenant/company or changing its business identity and branding. Company name and logo are returned by the company APIs and used by application branding; report-specific use depends on each report implementation.

Prerequisites: Authenticated administrator. No company dropdown or multi-company selector exists on this page.

Fields

Field	Technical mapping	UI/Required	Purpose and usage	Solar example	Validation / source / edit behavior
Company Code	<code>companyCode</code> → <code>companyCode</code> → <code>company_master.company_code</code>	Text; mandatory	Stable business code used to identify the company record.	SUNOPS	UI rejects blank; backend DTO is not blank; persisted code is unique. Editable when user can persist.
Company Name	<code>companyName</code> → <code>companyName</code> → <code>company_master.company_name</code>	Text; mandatory	Display/legal name used in company profile and branding.	Sunrise Solar Operations Pvt Ltd	UI rejects blank; backend not blank.
Email	<code>email</code> → <code>email</code> → <code>company_master.email</code>	Email; optional	General company contact.	operations@sunrise.example	Browser email control plus backend email validation where configured.
Phone Number	<code>phoneNumber</code> → <code>phoneNumber</code> → <code>company_master.phone_number</code>	Text; optional	Company contact number.	+91 98765 43210	No strict UI pattern; backend length applies.
Status	<code>status</code> → <code>status</code> → <code>company_master.status</code>	Dropdown; optional	Enables/inactivates the company profile.	ACTIVE	Static options <code>ACTIVE</code> , <code>INACTIVE</code> ; defaults to <code>ACTIVE</code> .
Address	<code>address</code> → <code>address</code> → <code>company_master.address</code>	Text area; optional	Registered/operational address used as company reference data.	Solar Park Road, Jaisalmer, Rajasthan	Free text.
Upload Logo	<code>multipart file</code> → stored logo metadata/path	File; optional	Replaces branding image.	sunrise-logo.png	Sent after company save to <code>/api/company/upload-logo</code> ; file rules come from backend multipart/allowed-content implementation. Existing preview remains until replaced.

Save, edit, cancel, APIs, and database

- Save calls `POST /api/company/create`; update calls `PUT /api/company/update/{id}`; logo upload follows with `POST /api/company/upload-logo`.
- The page remains on the company profile and reloads the saved response. Users without create/update permission see disabled fields.
- There is no destructive company delete action in the current UI/mapping.
- Main table: `company_master`; audit/status columns are defined in its Liquibase/entity mapping.

Tips: Use a stable short code and a landscape/transparent logo.

Common errors: blank code/name, invalid image, or `COMPANY_VIEW` without the appropriate save permission.

2. Site Create/Edit/View

Purpose, users, and timing

Navigation: HR & Sites → Sites → Add Site

Routes: `/hr/sites/new`, `/hr/sites/{id}/edit`, `/hr/sites/{id}/view`

Permissions: `SITE_VIEW`, `SITE_CREATE`, `SITE_UPDATE`, `SITE_DELETE`.

Creates the plant/site boundary used for equipment, employees, vendors, requests, downtime, inventory, reports, and record-level authorization.

Fields

Field	Technical mapping	UI/Required	Purpose and usage	Example	Validation / source / edit behavior
Site Code	<code>siteCode</code> → <code>siteCode</code> → <code>site_master.site_code</code>	Text; mandatory	Short unique plant identifier used in lists and references.	<code>RJ-SP-01</code>	Required; backend uniqueness/length applies. Editable on edit.
Site Name	<code>siteName</code> → <code>siteName</code> → <code>site_master.site_name</code>	Text; mandatory	Human-readable plant name.	Rajasthan Solar Park	Required.
Organization Name	<code>organizationName</code> → same → <code>organization_name</code>	Text; optional	Operating/legal organization associated with site.	Sunrise Solar O&M	Free text. No Company dropdown exists here.
Site Type	<code>siteType</code> → same → <code>site_type</code>	Text; optional	Classifies facility.	<code>SOLAR_POWER_PLANT</code>	Free text, not an enum/dropdown.
Status	<code>status</code> → same → <code>status</code>	Dropdown; optional	Controls active use and helper lists.	<code>ACTIVE</code>	Static <code>ACTIVE / INACTIVE</code> ; default active. Delete action marks inactive.
Address Line 1/2	<code>addressLine1</code> , <code>addressLine2</code> → same → corresponding columns	Text; optional	Physical site address.	Plot 14, Solar Park Road	Free text.
City/State/Country/Pincode	matching properties → matching columns	Text; optional	Geographic and postal reference.	Jaisalmer, Rajasthan, India, 345001	No dedicated country/state dropdown or pincode regex in UI.
Contact Person	<code>contactPerson</code> → same → <code>contact_person</code>	Text; optional	Primary site contact.	Anita Sharma	Free text.
Contact Mobile	<code>contactMobile</code> → same → <code>contact_mobile</code>	Text; optional	Site escalation number.	+91 99887 77665	No strict UI phone pattern.
Contact Email	<code>contactEmail</code> → same → <code>contact_email</code>	Email; optional	Site email contact.	<code>rjplant@sunrise.example</code>	Browser email control/backend validation.
Latitude/Longitude	<code>latitude</code> , <code>longitude</code> → same → matching columns	Number; optional	Map/geographic coordinates.	26.9157, 70.9083	Numeric UI; current page does not enforce geographic ranges.

Workflow and dependencies

1. Enter code/name and optional address/contact/location data.
2. Save with `POST /api/hr/sites` or update with `PUT /api/hr/sites/{id}`.
3. On success, return to `/hr/sites`.
4. The site becomes available through `GET /api/hr/sites` to equipment, employee, vendor, AMC, maintenance, downtime, spare, dashboard, and report pages, filtered by authorization where applicable.

Main table: `site_master`; primary key `site_id`. `DELETE /api/hr/sites/{id}` is exposed as “mark inactive,” preserving references. Cancel returns to the site list and discards unsaved changes without a custom confirmation.

Not implemented: Capacity and timezone fields requested in the expected specification are not present on the current site form/entity.

3. Employee, Login, Site, and Role Assignment Create/Edit

Purpose and prerequisites

Navigation: HR & Sites → Employees → Add Employee

Routes: `/hr/employees/new`, `/hr/employees/{id}/edit`, `/hr/employees/{id}/view`

Permissions: `EMPLOYEE_VIEW`, `EMPLOYEE_CREATE`, `EMPLOYEE_UPDATE`, `EMPLOYEE_DELETE`. Helper APIs for sites are mapped to the respective page permissions. Role loading also requires an allowed role API mapping; see inconsistencies.

Creates the employee master, optional login account, site responsibilities, and security-role assignments. At least one site assignment is mandatory. When login is enabled, username/password and at least one role assignment are required.

Identity and employment fields

Field	Technical mapping	UI/Required	Purpose / example	Validation and behavior
Employee Code	<code>employeeCode</code> → same → <code>employee_master.employee_code</code>	Text; mandatory	Stable workforce ID, e.g. <code>EMP-RJ-0042</code> .	UI required; backend unique/length rules.
First Name	<code>firstName</code> → same → <code>first_name</code>	Text; UI mandatory	Ravi	Required in UI/backend.
Last Name	<code>lastName</code> → same → <code>last_name</code>	Text; optional	Kumar	Optional.
Mobile Number	<code>mobileNumber</code> → same → <code>mobile_number</code>	Text; mandatory	Technician/contact number.	UI rejects blank; backend rules apply.
Email	<code>email</code> → same → <code>email</code>	Email; optional	<code>ravi.kumar@sunrise.example</code>	Email format where present; uniqueness may be enforced for linked users.
Gender	<code>gender</code> → same → <code>gender</code>	Dropdown; optional	MALE, FEMALE, OTHER	Static options.
Date of Birth	<code>dateOfBirth</code> → same → <code>date_of_birth</code>	Date; optional	1992-08-14	Date input; no age rule visible in UI.
Date of Joining	<code>dateOfJoining</code> → same → <code>date_of_joining</code>	Date; optional	2026-04-01	Date input.
Designation	<code>designation</code> → same → <code>designation</code>	Text; optional	Electrical Technician	Free text.
Department	<code>department</code> → same → <code>department</code>	Text; optional	Operations & Maintenance	Free text; no Team master exists.
Status	<code>status</code> → same → <code>status</code>	Dropdown; optional	ACTIVE	Static active/inactive; default active.

Login fields

Field	Technical mapping	UI/Required	Purpose / example	Validation and behavior
Enable Login	<code>loginEnabled</code>	Switch; optional	Creates/maintains a linked application user.	When false, login fields are disabled.
Username	<code>username</code> → linked user username	Text; conditional	<code>ravi.kumar</code>	Required when login enabled; uniqueness enforced by user service/database.
Auth Role	<code>authRole</code> → linked user base role	Dropdown; conditional	EMPLOYEE	Static authentication role option; not a substitute for permission-role assignments.
Account Status	<code>accountStatus</code> → linked user status	Dropdown; conditional	ACTIVE	Active/inactive options.
Reset Password	local UI flag	Checkbox; edit only	Reveals password controls for an existing linked account.	Not persisted as a database field.
Password / Confirm Password	<code>password</code> , <code>confirmPassword</code>	Password; conditional	Temporary secret supplied securely	Required for new login; values must match; never documented/screenshot with real secrets and never returned.

Site assignment rows

Field	Mapping	Required	Source / use
Site	siteAssignments[].siteId → employee_site_assignment.site_id	Yes	GET /api/hr/sites ; only accessible/non-inactive sites should be offered.
Role Name	siteAssignments[].roleName → role_name	Yes	Free text describing operational responsibility at that site.
Primary Site	primarySite → primary_site	One primary expected	Checkbox; selecting one clears other primary flags.
Effective From/To	matching properties → matching columns	Optional	Validity period for the site assignment.
Status	assignment status	Optional	Active/inactive static dropdown.

Permission-role assignment rows

Field	Mapping	Required	Source / use
Scope	scopeType / siteId	Conditional	Global or site-specific authorization scope.
Role	roleId	Required when login enabled	Loaded from role service; selects permissions applied to the account.
Status	status	Optional	Active/inactive role assignment.

Save uses POST /api/hr/employees ; update uses PUT /api/hr/employees/{id} ; success returns to employee list. Main tables include employee_master , employee_site_assignment , linked users , and user-role assignment tables. The service validates duplicate sites, date ranges, roles, password rules, and site access.

Common mistakes: enabling login without a password/role, duplicate site rows, no primary site, password mismatch, or assigning an inaccessible site.

4. Role Create/Edit and Permission Assignment

Navigation: Administration → Roles → Add Role

Routes: `/admin/roles/new`, `/admin/roles/{id}/edit`; view has a separate read-only page.

Permissions: `ROLE_VIEW`, `ROLE_CREATE`, `ROLE_UPDATE`, `ROLE_DELETE`; permission catalogue uses `PERMISSION_VIEW` APIs where mapped.

Roles group permissions that control frontend routes/buttons and backend API authorization.

Field	Mapping	UI/Required	Purpose / example	Validation/source/edit
Role Code	<code>roleCode</code> → same → role table <code>role_code</code>	Text; mandatory	Stable machine code, e.g. <code>SOLAR_TECHNICIAN</code> .	Required; normalized/unique by backend. Changing a used code can affect configuration references.
Role Name	<code>roleName</code> → same → <code>role_name</code>	Text; mandatory	Display name, e.g. Solar Technician.	Required.
Description	<code>description</code> → same → <code>description</code>	Text; optional	Explains intended responsibility.	Free text.
Status	<code>status</code> → same → <code>status</code>	Dropdown; mandatory	Activates/inactivates role.	Static <code>ACTIVE / INACTIVE</code> .
Permissions	<code>permissionIds[]</code> → role- permission join table	Grouped checkboxes; optional	Grants page/action/API abilities.	Loaded from <code>GET /api/admin/permissions</code> ; searchable and grouped. Existing selections load from role detail.

Save: `POST /api/admin/roles`; update: `PUT /api/admin/roles/{id}`; cancel/success returns to `/admin/roles`. Role changes affect users assigned that role and cached API-permission evaluation; backend change flows should clear relevant cache as implemented.

Tip: Grant the smallest permission set and include helper API permissions required by a page. **Mistake:** granting only a route permission while omitting dropdown/helper API access.

5. Standalone User Account Creation

Navigation/Route: `/create-user` (protected by `USER_ROLE_ASSIGN`)

Purpose: Creates an application user through the HR-oriented user service. This is separate from the richer employee form and can produce identity duplication if governance is unclear.

Field	Frontend/API property	UI/Required	Purpose/example	Validation/source
Username	<code>username</code>	Text; mandatory	Login ID, <code>plant.admin2</code> .	UI rejects blank; backend uniqueness.
Email	<code>email</code>	Email; mandatory	Account contact/login recovery identity.	UI required/email control; backend email/uniqueness.
First Name	<code>firstName</code>	Text; mandatory	Meera	UI required.
Last Name	<code>lastName</code>	Text; mandatory	Singh	UI required.
Department	<code>department</code>	Text; mandatory	Plant Administration	UI required.
Role	<code>role</code>	Dropdown; mandatory	Base role assigned to account.	Static page options; default must be reviewed against backend accepted values.
Manager ID	<code>managerId</code>	Number; optional	Links a manager by numeric user/employee identifier.	No dropdown; incorrect IDs may fail backend validation.

The page first resolves the current HR user context and calls the user creation service, ultimately using the mapped user API (`POST /api/users`). Success shows a temporary-password message returned by the server and redirects to the first permitted route. Never expose that temporary password in documentation or screenshots.

Implementation issue: This page uses direct MUI fields rather than project common form components and overlaps employee login creation.

6. Vendor Create/Edit and Site Assignment

Navigation: Vendors → Add Vendor

Routes: `/vendors/new`, `/vendors/{id}/edit`

Permissions: `VENDOR_VIEW`, `VENDOR_CREATE`, `VENDOR_UPDATE`, `VENDOR_DELETE`.

Used for maintenance partners, equipment/service suppliers, AMC vendors, and reorder sourcing. At least one site assignment is required.

Field	Mapping	UI/Required	Purpose/example	Validation/source/edit
Vendor Code	<code>vendorCode</code> → same → <code>vendor_master.vendor_code</code>	Text; mandatory	Stable ID, <code>INV-SVC-01</code> .	Required and unique.
Vendor Name	<code>vendorName</code> → same → <code>vendor_name</code>	Text; mandatory	Solar Inverter Services India.	Required.
Contact Person	<code>contactPerson</code> → same → <code>contact_person</code>	Text; optional	Amit Patel	Free text.
Email	<code>email</code> → same → <code>email</code>	Email; optional	<code>support@vendor.example</code>	Email format.
Phone	<code>phone</code> → same → <code>phone</code>	Text; optional	+91 90000 10001	Backend length/no strict UI pattern.
Service Category	<code>serviceCategory</code> → same → <code>service_category</code>	Text; optional	INVERTER_MAINTENANCE	Free text, not a managed services list.
Active Vendor	<code>active</code> → same → active/status column	Switch; optional	Controls selectable vendor availability.	Boolean default from form/entity.
Address	<code>address</code> → same → <code>address</code>	Text area; optional	Registered/service address.	Free text.
Assigned Site	<code>siteAssignments[].siteId</code> → vendor-site table	Dropdown; mandatory per row	Site where vendor may be selected.	<code>GET /api/hr/sites</code> ; duplicate/site-access validation.
Assignment Status	<code>row status</code>	Dropdown; optional	Activates vendor at that site.	Static active/inactive.

Save/update APIs are `POST /api/vendors` and `PUT /api/vendors/{id}`. Success returns to vendor list. Site assignment determines site-filtered vendor dropdowns in AMC, PM, and assignments. Main tables: `vendor_master` and vendor-site assignment table.

7. Vendor AMC Contract Create/Edit and Equipment Mapping

Navigation: Vendors → AMC Contracts → Create

Routes: `/vendor-amc/create`, `/vendor-amc/edit/{id}`

Permissions: `VENDOR_AMC_VIEW`, `VENDOR_AMC_CREATE`, `VENDOR_AMC_UPDATE`, `VENDOR_AMC_DELETE`, `VENDOR_AMC_ASSIGN_EQUIPMENT`, `VENDOR_AMC_RENEW`.

Creates a time-bound service contract, SLA, coverage, and equipment mappings used by maintenance requests and PM schedules.

Field	Mapping	UI/Required	Purpose/example	Validation/source/edit
Site	<code>siteId</code> → same → <code>vendor_amc_contract.site_id</code>	Dropdown; mandatory	Contract operating site.	GET <code>/api/hr/sites</code> ; locked on edit; filters vendors/equipment.
Vendor	<code>vendorId</code> → same → <code>vendor_id</code>	Dropdown; mandatory	Contracting service vendor.	GET <code>/api/vendors</code> , filtered to selected site assignments.
Contract Number	<code>contractNumber</code> → same → <code>contract_number</code>	Text; mandatory	AMC-INV-2026-001.	Required, unique.
Contract Name	<code>contractName</code> → same → <code>contract_name</code>	Text; mandatory	Inverter Comprehensive AMC FY26.	Required.
Contract Type	<code>contractType</code> → same → <code>contract_type</code>	Dropdown; optional	COMPREHENSIVE.	Static options from page constants.
Start/End Date	matching properties → matching columns	Dates; mandatory	2026-04-01 to 2027-03-31.	End must not precede start; coverage mappings must fit contract dates.
Status	<code>status</code> → same → <code>status</code>	Dropdown; optional	ACTIVE, DRAFT, etc.	Enum/static options; lifecycle rules enforced by service.
Contract Value	<code>contractValue</code> → same → <code>contract_value</code>	Number; optional	1250000.00.	Non-negative money.
Response Time	<code>responseTimeHours</code> → same → <code>response_time_hours</code>	Number; optional	Vendor acknowledgement SLA, e.g. 2.	Non-negative.
Resolution Time	<code>resolutionTimeHours</code> → same → <code>resolution_time_hours</code>	Number; optional	Restoration SLA, e.g. 8.	Non-negative.
Labor/Spares Included	<code>includesLabor</code> , <code>includesSpares</code> → matching columns	Yes/No dropdowns	Defines commercial coverage shown on request/AMC views.	Boolean.
Contact Person/Phone/Email	matching properties/columns	Inputs; optional	Contract-specific escalation contact.	Email control for email.
Covered Equipment	<code>equipmentIds[]</code> → <code>equipment_amc_mapping</code>	Multi-select; optional	Assets covered by contract.	Equipment loaded by API and filtered to selected site; overlapping active coverage is blocked.
Coverage Description	<code>coverageDescription</code> → same column	Text area; optional	Included services/components.	Free text.
Remarks	<code>remarks</code> → same column	Text area; optional	Administrative notes.	Free text.

Create calls POST `/api/vendor-amc` with equipment mappings. Edit calls PUT `/api/vendor-amc/{id}`, then synchronizes additions/removals using POST `/api/vendor-amc/{id}/equipment` and DELETE `/api/vendor-amc/{id}/equipment/{equipmentId}`. Success opens the AMC view page. Renewal uses a separate POST `/api/vendor-amc/{id}/renew` action and preserves historical contracts via `renewed_from_contract_id`.

8. Equipment Create/Edit

Navigation: Equipment → Add Equipment

Routes: /equipment/new, /equipment/{id}/edit

Permissions: EQUIPMENT_VIEW, EQUIPMENT_CREATE, EQUIPMENT_UPDATE, EQUIPMENT_DELETE.

Registers a maintainable asset used by requests, PM, downtime, AMC, spare BOM, documents, history, cost, and health calculations.

Field	Mapping	UI/Required	Purpose/example	Validation/source/edit
Equipment Code	equipmentCode → same → equipment_master.equipment_code	Text; mandatory	Unique asset ID, INV-B01-001.	Required/unique; service trims/validates.
Equipment Name	equipmentName → same → equipment_name	Text; mandatory	Central Inverter Block 01.	Required.
Site	siteId → same → site_id	Dropdown; mandatory	Ownership/security site.	GET /api/hr/sites; only active/accessible values.
Category	category → same → category	Text; mandatory	INVERTER.	Required free text; no category master page exists.
Location	location → same → location	Text; optional	Block 01 Inverter Room.	Free text.
Manufacturer	manufacturer → same → manufacturer	Text; optional	Sungrow.	Free text.
Model Number	modelNumber → same → model_number	Text; optional	SG3125HV.	Free text.
Serial Number	serialNumber → same → serial_number	Text; optional	Manufacturer serial.	Backend uniqueness if configured; optional UI.
Status	status → same → status	Dropdown	Master status.	ACTIVE, INACTIVE, UNDER_MAINTENANCE, RETIRED.
Lifecycle Status	lifecycleStatus → same → lifecycle_status	Dropdown	Asset lifecycle.	DRAFT, COMMISSIONED, ACTIVE, STANDBY, UNDER_MAINTENANCE, BREAKDOWN, DECOMMISSIONED, SCRAPPED.
Operating Status	operatingStatus → same → operating_status	Dropdown	Current operating state used on dashboard/health.	RUNNING, STANDBY, STOPPED, UNDER_MAINTENANCE, BREAKDOWN.
Asset Condition	assetCondition → same → asset_condition	Dropdown	Inspection/health condition.	GOOD, FAIR, POOR, CRITICAL, UNKNOWN.
Ownership Type	ownershipType → same → ownership_type	Dropdown	Commercial ownership.	OWNED, LEASED, RENTED, CUSTOMER_SUPPLIED.
Installation/Commissioning Date	matching properties/columns	Dates; optional	2024-02-15, 2024-03-01.	Service validates lifecycle chronology.
Warranty Expiry	warrantyExpiryDate → same → warranty_expiry_date	Date; optional	2029-02-14.	Date validation relative to installation where implemented.
Decommission Date	decommissionDate → same → decommission_date	Date; optional	Final service date.	Must align with lifecycle dates/status rules.
Criticality	criticality → same → criticality	Dropdown	Risk prioritization.	Static low/medium/high/critical; default medium.
Asset Number	assetNumber → same → asset_number	Text; optional	Finance asset ID.	Free text.
Purchase Date/Cost	matching properties → matching columns	Date/number; optional	2024-01-10, 8500000.	Cost non-negative; chronology validated.
Capitalization Date	capitalizationDate → same column	Date; optional	Finance capitalization date.	Finance-date validation.
Depreciation Method	depreciationMethod → same column	Dropdown; optional	STRAIGHT_LINE.	Static options; clearable.
Cost Center	costCenter → same column	Text; optional	RJ-INV-OPEX.	Used in cost grouping/report.
Department	department → same column	Text; optional	Electrical O&M.	Used in ownership/cost reporting.

Save calls `POST /api/equipment`; update calls `PUT /api/equipment/{id}`; success returns to equipment list. Equipment documents and spare BOM are separate embedded actions on the view page using `/documents` and `/spare-bom` APIs.

Not implemented on this form: equipment type master, parent equipment, vendor field, and meter configuration requested in the expected list.

9. Maintenance Request Create/Edit

Navigation: Maintenance → Requests → Add Request

Routes: `/maintenance/requests/new`, `/maintenance/requests/{id}/edit`

Permissions: `REQUEST_VIEW`, `REQUEST_CREATE`, `REQUEST_UPDATE`, `REQUEST_DELETE`; transitions use `REQUEST_UPDATE`.

Creates corrective, preventive, inspection, or calibration demand tied to a site/equipment. Requests feed assignment, downtime, AMC vendor handling, equipment history, approvals, and reports.

Field	Mapping	UI/Required	Purpose/example	Validation/source/edit
Site	<code>siteId</code> → same → <code>maintenance_request.site_id</code>	Dropdown; mandatory	Plant responsible for request.	<code>GET /api/hr/sites</code> ; filters equipment.
Equipment	<code>equipmentId</code> → same → <code>equipment_id</code>	Dropdown; mandatory	Affected asset.	<code>GET /api/equipment</code> ; filtered by selected site. Changing it rechecks AMC.
AMC information	read-only from active AMC	Information	Shows contract/vendor/SLA/coverage.	<code>GET /api/equipment/{id}/active-amc</code> .
Assign to AMC Vendor	<code>externalVendorAssignment</code>	Yes/No; conditional	Routes request context to covered vendor.	Shown only with active AMC; populates vendor/contract.
AMC Vendor	derived <code>vendorId</code>	Read-only	Displays selected AMC vendor.	Derived from active contract.
Vendor Reference Number	<code>vendorReferenceNumber</code> → same column	Text; conditional	Vendor ticket/case reference.	Shown only with AMC.
Title	<code>title</code> → same → <code>title</code>	Text; mandatory	Short issue, e.g. <code>Inverter communication failure</code> .	Required.
Reported By	<code>reportedBy</code> → same → <code>reported_by</code>	Text; optional	Person/source reporting issue.	Free text, not current-user dropdown.
Type	<code>requestType</code> → same → <code>request_type</code>	Dropdown	<code>BREAKDOWN</code> , <code>PREVENTIVE</code> , <code>INSPECTION</code> , <code>CALIBRATION</code> .	Defaults <code>BREAKDOWN</code> .
Priority	<code>priority</code> → same → <code>priority</code>	Dropdown	<code>LOW</code> , <code>MEDIUM</code> , <code>HIGH</code> , <code>URGENT</code> .	Defaults medium.
Requested Date	<code>requestedDate</code> → same → <code>requested_date</code>	Date; optional	Business request date.	Defaults today.
Target Completion	<code>targetCompletionDate</code> → same column	Date; optional	Due date used for overdue logic.	Should not precede request date; backend validation is authoritative.
Description	<code>description</code> → same → <code>description</code>	Text area; mandatory	Full symptoms/impact.	Required.

Create/update: `POST /api/maintenance/requests`, `PUT /api/maintenance/requests/{id}`. Success returns to request list. Status is excluded from normal edit payload; controlled workflow transitions use `POST /api/maintenance/requests/{id}/transition`. Request number, timestamps, approval metadata, and PM/AMC links are backend-generated/read-only.

Not implemented: failure-type dropdown and attachments on the request form.

10. Maintenance Assignment Create/Edit

Navigation: Maintenance → Assignments → Add Assignment

Routes: `/maintenance/assignments/new`, `/maintenance/assignments/{id}/edit`

Permissions: `ASSIGNMENT_VIEW`, `ASSIGNMENT_CREATE`, `ASSIGNMENT_UPDATE`, `ASSIGNMENT_DELETE`.

Field	Mapping	UI/Required	Purpose/example	Validation/source/edit
Site	<code>siteId</code> (derived through request)	Dropdown; mandatory	Filters requests, vendors, employees, and spares.	<code>GET /api/hr/sites</code> ; site access enforced.
Request	<code>requestId</code> → same → <code>maintenance_assignment.request_id</code>	Dropdown; mandatory	Work demand being assigned.	Maintenance request API, filtered by site.
Vendor	<code>vendorId</code> → same → <code>vendor_id</code>	Dropdown; optional	External assignee.	Vendors filtered by site.
Assigned Technician	<code>assignedEmployeeId</code> → same → <code>assigned_employee_id</code>	Dropdown; optional	Internal technician.	Employee search filtered to selected site/active records.
Assigned To	<code>assignedTo</code> → same → <code>assigned_to</code>	Text; mandatory	Display assignee; auto-filled/locked when technician selected.	Required.
Assigned Date	<code>assignedDate</code> → same → <code>assigned_date</code>	Date; optional	Assignment date; defaults today.	Backend fills today if omitted.
Planned Start/End	matching properties/columns	Dates; optional	Scheduled work window.	End must not precede start.
Status	<code>status</code> → same → <code>status</code>	Dropdown	Assignment lifecycle, default ASSIGNED.	Actual allowed statuses come from page/service rules.
Actual Start/End	matching properties/columns	Dates; optional	Actual execution dates.	End must not precede start.
Estimated Cost	<code>estimatedCost</code> → same → <code>estimated_cost</code>	Number; optional	Planned service cost.	Non-negative.
Service/Vendor Cost	<code>actualCost</code> → same → <code>actual_cost</code>	Number; optional	Actual non-material cost.	Non-negative.
Material Cost	calculated from spare usage	Read-only	Sum of qualifying consumed material.	Not editable.
Total Actual Cost	service + material	Read-only	Total assignment cost.	Calculated UI value.
Remarks	<code>remarks</code> → same → <code>remarks</code>	Text area; optional	Instructions/context.	Free text.

Save/update uses `POST /api/maintenance/assignments` and `PUT /api/maintenance/assignments/{id}`. After an ID exists, Checklist, Work Logs, and Spare Parts tabs become available.

11. Assignment Checklist Add/Edit/Proof Upload

Location: Assignment edit/view → Checklist tab

Permissions: ASSIGNMENT_CHECKLIST_VIEW, ASSIGNMENT_CHECKLIST_UPDATE ; the UI refers to proof-specific permission names, while CSV maps proof upload/delete under ASSIGNMENT_CHECKLIST_UPDATE —see inconsistencies.

Field	Mapping	Required	Purpose/validation
Task	taskTitle → checklist entity task_title	Required for meaningful row	Work step, e.g. Inspect DC terminal torque .
Instructions	instructions → instructions	No	Execution guidance.
Response Type	responseType → response_type	Default checkbox	CHECKBOX , TEXT , NUMBER , PHOTO .
Required	required → required	Boolean	Required steps can block assignment completion.
Proof Required	proofRequired → proof_required	Boolean	Requires file before completion when configuration enables it.
Status	row status	Yes during execution	PENDING , COMPLETED , NOT_APPLICABLE .
Reading/Response	responseValue → response_value	Conditional	Numeric/text/check response.
Remarks	remarks → remarks	No	Technician evidence/context.
Proof File	multipart file → checklist proof table	Conditional	Photo/PDF evidence; content type and 10 MB configured maximum.

APIs: GET/POST /api/maintenance/assignments/{assignmentId}/checklist ; PUT/DELETE row path; POST/DELETE proof paths. PM-generated assignments copy checklist template items, which cannot be deleted like manually added rows. Required/proof rules are enforced before completion.

12. Technician Work Log Create/Edit and Attachment Upload

Location: Assignment edit/view → Work Logs tab

Permissions: ASSIGNMENT_WORK_LOG_VIEW, ASSIGNMENT_WORK_LOG_CREATE, ASSIGNMENT_WORK_LOG_UPDATE, ASSIGNMENT_WORK_LOG_DELETE.

Field	Mapping	UI/Required	Purpose/example	Validation/source
Technician	technicianEmployeeId → technician_employee_id	Dropdown; mandatory	Person performing work.	Active employees from assignment site.
Start Time	startTime → start_time	Date-time; mandatory	Work start.	Required.
End Time	endTime → end_time	Date-time; optional	Work finish.	Must not precede start.
Status	completionStatus → completion_status	Dropdown	IN_PROGRESS, COMPLETED, FOLLOW_UP_REQUIRED, CANCELLED.	Defaults in progress.
Work Notes	workNotes → work_notes	Text; optional	Activity summary.	Free text.
Issue Found	issueFound → issue_found	Text; optional	Diagnosis, e.g. Cooling fan bearing seized.	Free text.
Action Taken	actionTaken → action_taken	Text; optional	Repair performed.	Free text.
Attachment	multipart file → work-log attachment table	Optional	Photo/report evidence.	Backend file limits/content validation.

APIs: POST /api/maintenance/assignments/{id}/work-logs; PUT/DELETE row paths; POST/DELETE attachment paths; GET/download operations require view authorization. Completion progress appears on the assignment.

13. Equipment Downtime Create/Edit

Navigation: Maintenance → Downtime → Add Downtime

Routes: /maintenance/downtime/new, /maintenance/downtime/{id}/edit

Permissions: DOWNTIME_VIEW, DOWNTIME_CREATE, DOWNTIME_UPDATE, DOWNTIME_DELETE plus special workflow permissions.

Field	Mapping	UI/Required	Purpose/example	Validation/source
Site	siteId → same → equipment_downtime.site_id	Dropdown; mandatory	Downtime site.	Site API; filters equipment.
Equipment	equipmentId → same → equipment_id	Dropdown; mandatory	Failed/affected asset.	Equipment filtered by site.
Maintenance Request	requestId → same → request_id	Dropdown; optional	Links corrective request.	Requests filtered by site/equipment.
Downtime Type	planned → same → planned	Dropdown	Planned/unplanned flag.	Boolean options.
Reason Category	reasonCategory → same → reason_category	Dropdown; optional	Standard cause group.	Static service-approved categories.
Reason Code	reasonCode → same → reason_code	Text; optional	Local code.	Free text.
Reason	reason → same → reason	Text; mandatory	Immediate reason, e.g. Transformer oil leakage.	Required.
Start/End Time	downtimeStart, downtimeEnd → matching columns	Date-time; start mandatory	Downtime interval.	End must be after start; overlapping records are rejected.
Production Line/Shift/Operator	matching properties/columns	Text; optional	Operational context.	Free text.
Expected Output / Hour	expectedOutputPerHour → same column	Number; optional	Expected production rate.	Non-negative.
Loss Rate / Unit	lossRatePerUnit → same column	Number; optional	Monetary value per lost unit.	Non-negative.
Lost Quantity/Minutes/Hours/Lost Amount	calculated entity/UI values	Read-only	Production and duration impact.	Derived from interval/rates.
Root Cause	rootCause → same → root_cause	Text area; optional initially	Confirmed cause.	Required before closing major downtime.
Remarks	remarks → same → remarks	Text area; optional	Operational notes.	Free text.

Create/update APIs: POST /api/maintenance/downtime, PUT /api/maintenance/downtime/{id}. Status starts OPEN unless valid initial state. Workflow actions: confirm, start maintenance, restore, verify, close, and reopen use their specific POST APIs/permissions. Main table equipment_downtime stores loss, verification, closure, and audit fields.

14. Downtime RCA Action Add/Edit

Location: Downtime view → RCA Actions

Permission: DOWNTIME_RCA_MANAGE ; unavailable after closed/cancelled.

Field	Mapping	Required	Purpose/validation
Action Type	actionType → RCA action column	No/default	Corrective/preventive action classification from static options.
Target Date	targetDate → target date column	Optional	Due date for RCA action.
Status	status → status column	Default	RCA action progress from static options.
Action Description	description → description column	Mandatory	Specific corrective/preventive action.

POST /api/maintenance/downtime/{id}/rca-actions ; PUT row path. Major downtime cannot close without a root cause and at least one RCA action. Timeline/history records transitions and action changes.

15. Preventive Maintenance Schedule and Checklist Create/Edit

Navigation: Maintenance → Preventive Maintenance → Add Schedule

Routes: `/maintenance/preventive/new`, `/maintenance/preventive/{id}/edit`

Permissions: currently reused `REQUEST_VIEW`, `REQUEST_CREATE`, `REQUEST_UPDATE`, `REQUEST_DELETE`; calendar has `PM_CALENDAR_VIEW`.

Field	Mapping	UI/Required	Purpose/example	Validation/source
Site	<code>siteId</code> → same → PM table <code>site_id</code>	Dropdown; mandatory	Plant/site.	Site API; filters equipment/vendors.
Equipment	<code>equipmentId</code> → same → <code>equipment_id</code>	Dropdown; mandatory	Asset maintained.	Equipment filtered by site.
AMC Coverage	<code>amcContractId</code> → same → <code>amc_contract_id</code>	Dropdown; optional	Contract covering PM.	Active AMC API; disabled if none; dates must be covered.
Assigned Vendor	<code>vendorId</code> → same → <code>vendor_id</code>	Dropdown; optional	External PM performer.	Vendors assigned to site.
Assigned To	<code>assignedTo</code> → same → <code>assigned_to</code>	Text; optional	Named internal/external assignee.	Free text.
PM Task	<code>title</code> → same → <code>title</code>	Text; mandatory	Quarterly inverter cooling inspection.	Required.
Frequency	<code>frequency</code> → same → <code>frequency</code>	Dropdown; mandatory	Recurrence.	Static supported frequencies; default monthly.
Priority	<code>priority</code> → same → <code>priority</code>	Dropdown	Risk priority.	Default medium.
Status	UI <code>active</code> → entity <code>active</code>	Dropdown	Enables recurrence.	Boolean active/inactive options.
Approval Status	<code>status</code> → same → <code>status</code>	Dropdown	Approval/lifecycle state.	Creation/update can become pending approval according to configuration.
Start/End/Next Due Date	matching properties/columns	Dates; start/next mandatory	Recurrence boundaries.	Next due cannot precede start or exceed end; end cannot precede start.
Description	<code>description</code> → same → <code>description</code>	Text area; mandatory	Work scope.	Required.

Checklist template fields: Step/task title (mandatory), Instructions, Response (`CHECKBOX`, `TEXT`, `NUMBER`, `PHOTO`), Required, Proof Required, Active, and sequence controlled by move-up/down. Template rows are copied to generated assignments.

Create/update: `POST /api/preventive-maintenance/schedules`, `PUT /api/preventive-maintenance/schedules/{id}`.
Work-order generation uses POST schedule action APIs and creates a maintenance request, optional assignment, and copied checklist. Schedule code is generated when absent and is not entered on the current UI.

Not implemented: runtime threshold/meter-triggered PM.

16. Spare Part/Site Stock Create/Edit and Equipment BOM Link

Navigation: Inventory → Spare Parts → Add Spare Part

Routes: `/inventory/spare-parts/new`, `/inventory/spare-parts/{id}/edit`

Permissions: `SPARE_PART_VIEW`, `SPARE_PART_CREATE`, `SPARE_PART_UPDATE`, `SPARE_PART_DELETE`.

Field	Mapping	UI/Required	Purpose/example	Validation/source/edit
Part Code	<code>partCode</code> → same → <code>spare_part_master.part_code</code>	Text; mandatory	Unique code, <code>IGBT-1700V-01</code> .	Required/unique; locked on edit.
Part Name	<code>partName</code> → same → <code>part_name</code>	Text; mandatory	<code>1700V IGBT Module</code> .	Required; shared master value across sites.
Unit	<code>unit</code> → same → <code>unit</code>	Text; mandatory	<code>EA</code> , <code>L</code> , <code>M</code> .	Required free text.
Category	<code>category</code> → same → <code>category</code>	Text; optional	<code>INVERTER_ELECTRONICS</code> .	No category master.
Site	<code>siteId</code> → same → <code>spare_part_site_stock.site_id</code>	Dropdown; mandatory	Stock-owning site.	Site API; locked on edit.
Preferred Vendor	<code>preferredVendorId</code> → same → master vendor FK	Dropdown; optional	Suggested reorder supplier.	Vendors filtered to selected site.
Opening Stock	<code>currentStock</code> → same → <code>current_stock</code>	Number; create only	Initial physical balance.	Non-negative; creates <code>OPENING_BALANCE</code> transaction; locked on edit.
Minimum Stock	<code>minimumStock</code> → same → <code>minimum_stock</code>	Number; optional	Low-stock threshold.	Non-negative.
Unit Cost	<code>unitCost</code> → same → <code>unit_cost</code>	Number; optional	Inventory/usage valuation.	Non-negative.
Status	<code>status</code> → stock status	Dropdown	Site-stock availability.	Active/inactive.
Storage Location	<code>storageLocation</code> → same column	Text; optional	<code>Main Store/Rack A/Bin 03</code> .	Free text.
Description	<code>description</code> → master description	Text area; optional	Part specification.	Free text.

Equipment BOM link fields: Equipment (mandatory, filtered by site), Recommended Qty (positive mandatory), Criticality, Replacement Frequency, Status, Remarks. Create/update/delete uses `/api/spare-parts/{stockId}/equipment-bom` APIs. The link associates the global part and site stock with compatible equipment.

Save uses `POST /api/spare-parts`; edit `PUT /api/spare-parts/{stockId}`. Main tables: `spare_part_master`, `spare_part_site_stock`, `equipment_spare_bom`, and `spare_part_transaction` for opening balance.

17. Stock In, Adjustment, Transfer, and Import

Location: Inventory → Spare Parts list action dialogs

Permissions: STOCK_TRANSACTION_CREATE ; import uses SPARE_PART_CREATE ; history uses STOCK_TRANSACTION_VIEW .

Stock In / Adjustment fields

Field	API property	Required	Rules/use
Quantity to Add / New Stock Quantity	quantity	Yes	Stock-in must be positive; adjustment is a non-negative target balance and cannot be below reserved stock.
Unit Cost	unitCost	Optional	Defaults to current cost; non-negative. Stock-in currently replaces current site cost.
Remarks	remarks	Optional	Audit explanation. Strongly recommended for adjustment.

APIs: POST /api/spare-parts/{stockId}/stock-in or /adjust . Both lock stock, update balance, write before/after transaction, and may trigger low-stock notification.

Site-to-site transfer fields

Field	API property	Required	Rules/use
Target Site	targetSiteId	Yes	Site API; excludes source; user needs access to both sites.
Quantity	quantity	Yes	Positive and cannot exceed available (current-reserved).
Target Storage Location	targetStorageLocation	Optional	Creates/updates destination location.
Remarks	remarks	Optional	Transfer audit note.

POST /api/spare-parts/{sourceStockId}/transfer immediately writes TRANSFER_OUT and TRANSFER_IN ; there is no dispatch/receipt in-transit workflow.

Import

Accepts .xlsx , .xls , or .csv at POST /api/spare-parts/import ; returns created/updated/failed counts and row errors. Import can create opening/adjustment transactions. Use sanitized templates and review site IDs/codes before upload.

18. Spare Request, Approval, Store Fulfilment, Consumption, and Return

Request

Location: Assignment edit → Spare Parts tab

Permissions: SPARE_USAGE_VIEW, SPARE_USAGE_CREATE, SPARE_USAGE_UPDATE, SPARE_USAGE_DELETE.

Field	Mapping	Required	Source/use
Spare Part	stockId → site stock FK	Yes by backend	Site stock API, filtered to assignment site; equipment BOM recommendations may be loaded.
Quantity / Requested Quantity	quantityUsed	Positive	Requested amount. One part/site-stock row per assignment due to unique constraint.
Remarks	remarks	Optional	Need/reason/instructions.

POST `/api/maintenance/assignments/{id}/spares` (or `/api/assignments/{id}/spare-requests` in the store-oriented flow).
Initial status REQUESTED.

Manager approval

Route: `/inventory/spare-approvals`; permission SPARE_USAGE_MANAGER_APPROVE. Fields: Approved Qty (positive, cannot exceed business limits enforced by service) and Remarks. POST manager approve/reject endpoints.

Store processing

Route: `/inventory/spare-requests`. Actions and permissions:

- Check stock: SPARE_USAGE_STORE_PROCESS.
- Reserve: SPARE_USAGE_RESERVE.
- Issue: SPARE_USAGE_ISSUE.
- Create purchase request when unavailable: REORDER_CREATE.
- Consume/return: SPARE_USAGE_CONSUME.

Consume/return fields are Issued Qty (read-only), Consumed Qty, Returned Qty, and Remarks. Consumed plus returned cannot exceed issued; returns restore stock and consumption sets material cost.

```
REQUESTED → MANAGER_APPROVED → STORE_REVIEW
→ STOCK_AVAILABLE → RESERVED → ISSUED → CONSUMED/RETURNED
→ STOCK_NOT_AVAILABLE → PURCHASE_REQUESTED → PURCHASE_RECEIVED → reserve/issue
```

Main table: `maintenance_spare_usage`; links assignment, stock, part, users performing each action, timestamps, quantities, costs, and reorder reference.

19. Reorder/Purchase Request Create, Edit, and Receipt

Navigation: Inventory → Reorders; creation is also available from the spare list or stock-shortage flow.

Permissions: REORDER_VIEW, REORDER_CREATE, REORDER_UPDATE.

Field	Mapping	Required	Purpose/validation
Stock/Part/Site	derived from selected stock	System	Links purchase need to exact site inventory.
Requested Quantity	requestedQuantity → reorder table column	Positive mandatory	Quantity to source.
Estimated Unit Cost	estimatedUnitCost → matching column	Non-negative	Defaults to stock unit cost.
Estimated Total Cost	calculated	Read-only	Quantity × estimated unit cost.
Vendor	vendorId → vendor FK	Optional	Preferred/selected vendor must be assigned to site. Creation dialog currently defaults through service and does not expose vendor selection.
Expected Date	expectedDate → matching column	Optional	Planned arrival.
Status	status → status column	Default/request edit	Current reorder lifecycle; edit dialog exposes service-supported statuses.
Remarks	remarks	Optional	Purchase justification/context.

Create: POST /api/spare-part-reorders; edit: PUT /api/spare-part-reorders/{id}. Receipt dialog captures Received Quantity, Unit Cost, and Remarks and calls POST /api/spare-part-reorders/{id}/receive-stock, which posts stock-in and marks the request received.

Current limitation: any positive receipt marks the entire reorder RECEIVED; partial/multiple receiving is not modeled.

20. Approval Configuration and Approval Decisions

Approval configuration

Route: /admin/approval-config

Permissions: APPROVAL_CONFIG_VIEW, APPROVAL_CONFIG_UPDATE .

Field	Mapping	Required	Source/use
Module	moduleCode	Read-only in edit	Business module/action owner. Existing rows provide value.
Action	actionCode	Read-only in edit	Operation requiring approval.
Approval Required	approvalRequired	Switch	Enables workflow interception.
Approver Role	approverRoleCode	Conditional	Roles API; recipients must hold role/scope.
Minimum Approval Count	minApprovalCount	Minimum 1	Number of approvals needed.
Status	status	Default active	Active/inactive config.

GET/POST /api/admin/approval-config; PUT row path. The current dialog primarily edits seeded rows; create support exists in service but UI entry begins from a selected row.

Approve/reject

Route: /approvals/pending; permissions APPROVAL_VIEW, APPROVAL_APPROVE, APPROVAL_REJECT . Decision form contains Comments; submit uses POST /api/approvals/{id}/approve or /reject . Approval result invokes the registered business transition. History is read-only/searchable.

21. Notification Configuration

Route: `/admin/notification-settings`

Permissions: `NOTIFICATION_CONFIG_VIEW`, `NOTIFICATION_CONFIG_UPDATE`.

Field	Mapping	Required/default	Purpose/validation
Notifications Enabled	<code>enabled</code>	Boolean	Global notification switch.
In-App Notifications	<code>inAppEnabled</code>	Boolean	Enables notification-center messages.
Email Notifications	<code>emailEnabled</code>	Boolean	Enables email channel; SMTP must be configured.
PM Due Reminders	<code>pmDueReminderEnabled</code>	Boolean	Sends upcoming-PM alerts.
Overdue Request Alerts	<code>overdueRequestEnabled</code>	Boolean	Sends overdue request alerts.
Approval Pending Alerts	<code>approvalPendingEnabled</code>	Boolean	Sends pending approval alerts.
PM Reminder Days	<code>pmReminderDays</code>	Number	Lead days; non-negative/service validated.
Daily Scan Time	<code>dailyScanTime</code>	Time	UI configuration value for scan time; deployment cron behavior must be verified when changed.
PM Recipient Roles	<code>pmRecipientRoleCodes[]</code>	Multi-select	Roles receiving PM reminders.
Overdue Recipient Roles	<code>overdueRecipientRoleCodes[]</code>	Multi-select	Roles receiving overdue alerts.
Approval Fallback Roles	<code>approvalFallbackRoleCodes[]</code>	Multi-select	Recipients when configured approver cannot be resolved.

GET/PUT `/api/admin/notification-settings`. Role options come from role service. Save remains on page and refreshes returned settings. Settings table/entity stores configuration; application properties provide defaults.

22. Permission Matrix

Page/workflow	View	Create	Update	Delete	Special actions
Company	COMPANY_VIEW	COMPANY_CREATE	COMPANY_UPDATE	—	Logo: COMPANY_UPDATE
Site	SITE_VIEW	SITE_CREATE	SITE_UPDATE	SITE_DELETE	Delete marks inactive
Employee	EMPLOYEE_VIEW	EMPLOYEE_CREATE	EMPLOYEE_UPDATE	EMPLOYEE_DELETE	Login/roles included in save
Role	ROLE_VIEW	ROLE_CREATE	ROLE_UPDATE	ROLE_DELETE	Catalogue: PERMISSION_VIEW
User account	USER_ROLE_VIEW	USER_ROLE_ASSIGN	USER_ROLE_UPDATE	—	Role assignment API
Vendor	VENDOR_VIEW	VENDOR_CREATE	VENDOR_UPDATE	VENDOR_DELETE	Site assignment included
AMC	VENDOR_AMC_VIEW	VENDOR_AMC_CREATE	VENDOR_AMC_UPDATE	VENDOR_AMC_DELETE	Assign equipment, renew
Equipment	EQUIPMENT_VIEW	EQUIPMENT_CREATE	EQUIPMENT_UPDATE	EQUIPMENT_DELETE	Documents/BOM mapped to update/delete
Request/PM	REQUEST_VIEW	REQUEST_CREATE	REQUEST_UPDATE	REQUEST_DELETE	PM calendar PM_CALENDAR_VIEW
Assignment	ASSIGNMENT_VIEW	ASSIGNMENT_CREATE	ASSIGNMENT_UPDATE	ASSIGNMENT_DELETE	Checklist/work-log permissions
Downtime	DOWNTIME_VIEW	DOWNTIME_CREATE	DOWNTIME_UPDATE	DOWNTIME_DELETE	Confirm, verify, close, reopen, RCA
Spare master	SPARE_PART_VIEW	SPARE_PART_CREATE	SPARE_PART_UPDATE	SPARE_PART_DELETE	Stock transaction permissions
Spare usage	SPARE_USAGE_VIEW	SPARE_USAGE_CREATE	SPARE_USAGE_UPDATE	SPARE_USAGE_DELETE	Approve, reserve, issue, consume, return, reject, cancel
Reorder	REORDER_VIEW	REORDER_CREATE	REORDER_UPDATE	—	Receipt uses update
Approval config	APPROVAL_CONFIG_VIEW	APPROVAL_CONFIG_UPDATE	APPROVAL_CONFIG_UPDATE	—	Decisions use approve/reject
Notification config	NOTIFICATION_CONFIG_VIEW	—	NOTIFICATION_CONFIG_UPDATE	—	—

23. API Dependency Summary

The authoritative row-level matrix is [API Dependency Matrix](#). Important patterns:

- Create/edit pages require their write API plus all site/equipment/vendor/employee/role helper GET APIs.
- Backend paths in the permission CSV include `/api`, matching the configured servlet context.
- `JwtFilter` / `ApiPermissionService` owns API permission authorization; services own record/site validation.
- Normal JSON endpoints return `ApiResponse<T>`; binary downloads are protocol exceptions.

24. Database Reference Summary

The authoritative page/table matrix is [Database Reference](#). Core tables include:

- company_master, site_master
- employee, employee-site, user, role, permission, and role-assignment tables
- vendor_master, vendor-site assignment, vendor_amc_contract, equipment_amc_mapping
- equipment_master, equipment documents, equipment_spare_bom
- maintenance_request, maintenance_assignment, checklist/work-log and attachment tables
- equipment_downtime, downtime history, and RCA action tables
- preventive_maintenance_schedule and PM checklist items
- spare_part_master, spare_part_site_stock, spare_part_transaction
- maintenance_spare_usage, spare_part_reorder_request
- approval configuration/request/history and notification settings tables

25. Workflow Linkages

Equipment maintenance

Site → Equipment → PM Schedule / Maintenance Request
→ Assignment → Checklist + Work Log + Spare Request
→ Downtime / Cost / Equipment History / Reports

Authorization

Employee/Login → Role Assignment → Permission
→ Frontend Route/Button → API Permission Mapping
→ JwtFilter/API Permission Service → Site/record access

Inventory

Spare Master + Site Stock → Assignment Spare Request
→ Manager Approval → Store Check → Reserve → Issue
→ Consume or Return
 ↳ shortage → Reorder → Receipt → Reserve/Issue

AMC

Vendor + Site → AMC Contract → Equipment Mapping
→ Active AMC lookup in Request/PM → Vendor context and SLA
→ Renewal preserves expired contract history

26. Expected Pages Not Currently Implemented

Expected page	Current finding
Team Create/Edit	No team route, form, controller, entity, or Liquibase table found. Department/site-role text is used instead.
Equipment Category/Type masters	Category is free text on equipment; no standalone category/type pages.
Meter Reading Create/Correction	No route/module/entity/API found.
User Profile Update	Navbar displays identity but has no profile edit route.
Change Password	Navbar links to <code>/change-password</code> , but <code>App.jsx</code> has no matching route/page; link falls through wildcard navigation.
User Role Assignment editor	<code>/admin/user-roles</code> exists but renders only an informational placeholder. APIs exist.
Report Configuration	Reports have runtime filters only; no saved report configuration form.
Store/Warehouse master	Storage location is free text; no store/bin master.
Purchase Order/quotation	Reorder request exists; full procurement workflow does not.
Request attachments	Not present on request creation; attachments exist for work logs/checklist/equipment documents.
PM runtime/meter trigger	Calendar/date frequency exists; runtime threshold/meter type does not.

27. Inconsistencies and Implementation Issues Found

1. Navbar change-password link has no route/page.
2. User Role Assignment page is a placeholder despite available APIs.
3. Employee role dropdown API access is not clearly mapped under `EMPLOYEE_CREATE/UPDATE` in the CSV; test runtime with restriction enabled.
4. Role form loads the permission catalogue, but `ROLE_CREATE/UPDATE` do not have explicit `GET /api/admin/permissions` helper rows; users may also need `PERMISSION_VIEW`.
5. Notification settings load roles, but notification configuration permissions do not explicitly map role helper APIs.
6. Assignment UI checks proof/attachment-specific permission names (`ASSIGNMENT_CHECKLIST_PROOF_UPLOAD`, etc.) while the CSV maps those APIs under broader checklist/work-log permissions. Button visibility and API authorization can diverge.
7. PM routes reuse maintenance-request permission codes instead of dedicated PM create/update/view permissions.
8. User Management and several dialog pages directly use MUI form fields rather than required common components.
9. Site expected capacity/timezone fields are absent.
10. Equipment expected type/vendor/parent/meter configuration fields are absent.
11. Maintenance request expected failure type and attachments are absent.
12. Reorder receipt marks the full request received after any positive receipt; partial receipts are not modeled.
13. Spare part master and site stock fields share one edit form; editing one site's record can update shared master details.
14. Transfer posts source-out/destination-in immediately; no in-transit/receipt confirmation.
15. API restriction is currently configured `cmms.security.api-permission-restriction-enabled=false`; production should deliberately review this setting.

28. Common Validation and Troubleshooting

- A 401 means authentication expired/missing; sign in again.
- A 403 usually means missing permission or site scope; compare route permission and helper API mapping.
- A 400 validation response should show `code`, `message`, `details[]`, and `correlationId`; correct the named field.
- “Unable to load form data” often indicates a blocked helper API, not a save failure.
- Date end values must not precede corresponding start values.
- Quantity/cost fields must be positive or non-negative as described.
- Unique business codes should be stable and should not be recycled.
- Do not enter real passwords in demos, screenshots, tickets, or documents.

29. Screenshot Register

Screenshots are stored under `documentation/screenshots/`. Capture status is recorded in [Screenshot Register](#). Figures must contain sanitized demo data and no tokens/passwords.

30. Glossary

Term	Meaning
AMC	Annual Maintenance Contract
BOM	Bill of Materials/compatible spare list
CMMS	Computerized Maintenance Management System
PM	Preventive Maintenance
RCA	Root Cause Analysis
SLA	Service Level Agreement
Site scope	Set of plant sites a user is authorized to access
Reservation	Stock quantity committed but not yet issued

31. Verification Checklist

- ☒ All routes in `App.jsx` reviewed.
- ☒ All standalone create/edit/configuration forms reviewed.
- ☒ Embedded checklist, work-log, downtime-RCA, stock, transfer, reorder, approval, issue, consume, return, import, and upload workflows reviewed.
- ☒ Frontend fields checked against DTO/entity/service behavior.
- ☒ Permission CSV reviewed.
- ☒ Missing expected pages identified without inventing them.
- ☐ Authenticated screenshots: dependent on runnable local services/demo authentication; see screenshot register.
- ☒ Markdown, Word, and PDF deliverables generated and validated during documentation build.